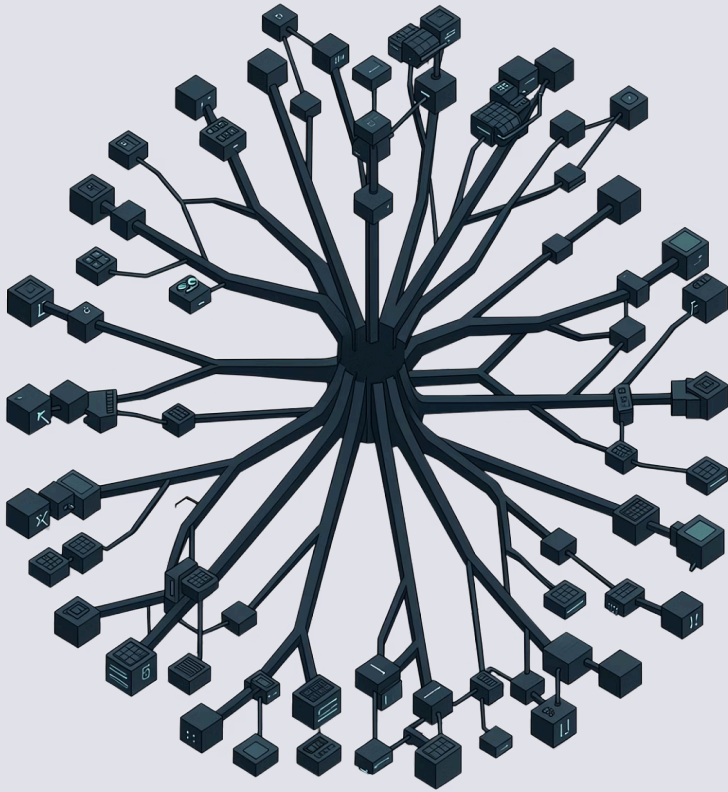




Hierarchical Clustering

A Step-by-Step Guide for B.Tech Students

Dr. Anilkumar V. Brahmane | Computer Engineering | Sanjivani College of Engineering

What is Clustering?



Definition

Grouping similar data points together so that points in the same group (cluster) are more similar to each other than to those in other groups. Hierarchical clustering is a powerful unsupervised learning technique that builds a hierarchy of clusters.

Why Hierarchical?

Unlike K-Means, you do **NOT** need to specify the number of clusters in advance — just run once and cut the tree at any level. The algorithm iteratively merges or splits groups based on a distance measure.

Output

A tree-like diagram called a **Dendrogram** that shows how clusters are formed step by step — one run, infinite clustering options.

Two Approaches at a Glance

▲ Agglomerative (Bottom-Up)

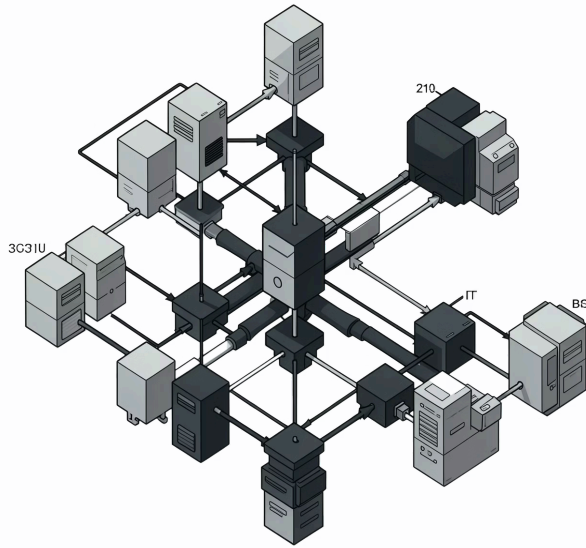
Start with every data point as its own cluster → keep merging the two closest clusters → stop when all points form one big cluster.

▼ Divisive (Top-Down)

Start with all data points in one cluster → keep splitting into smaller clusters → stop when every point is its own cluster.

Feature	Agglomerative	Divisive
Direction	Bottom → Top	Top → Bottom
Start	Each point = 1 cluster	All points = 1 cluster
Process	Merge closest clusters	Split largest cluster
Complexity	$O(n^2)$	$O(2^n)$ – more complex
Common Use	Most widely used	Large distinct groups

Agglomerative Clustering — Step by Step



01

Initialize

Treat each of the N data points as a separate cluster — N clusters total.

02

Compute Distance

Calculate the distance (e.g., Euclidean) between every pair of clusters.

03

Merge

Combine the two closest clusters into one new cluster.

04

Update & Repeat

Recalculate distances and go back to Step 2 until only 1 cluster remains.

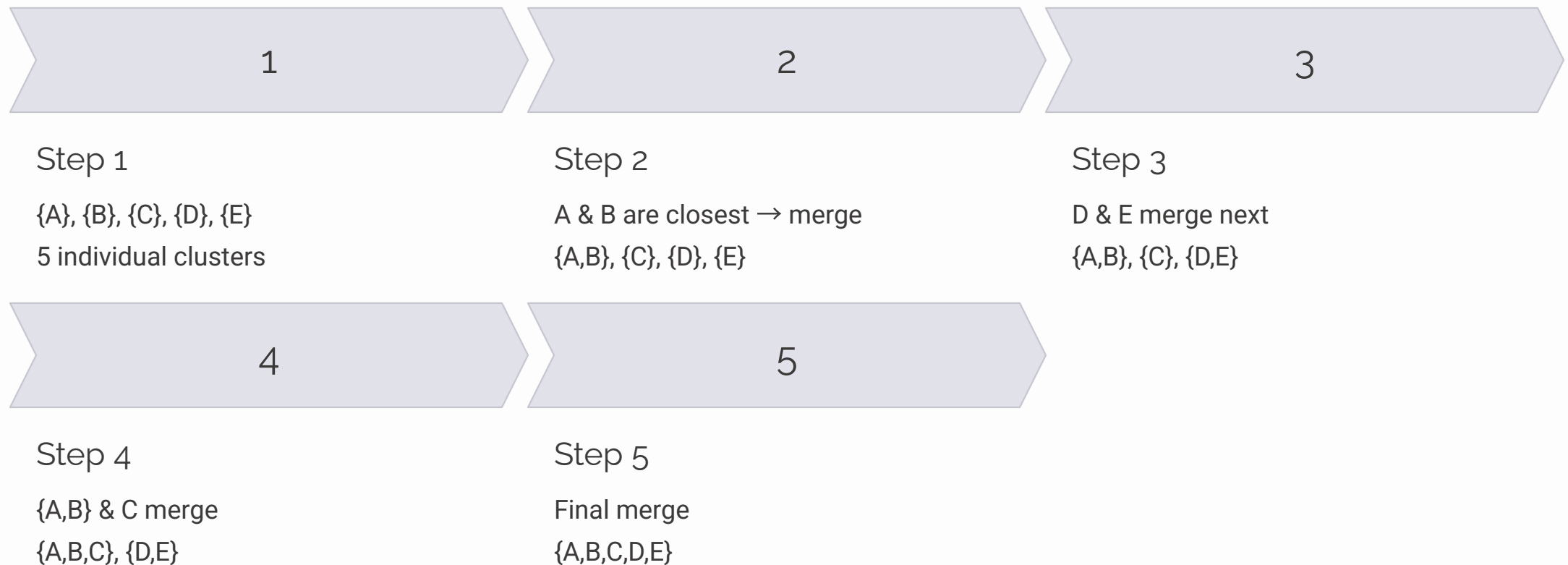
05

Cut Dendrogram

Choose a threshold height to extract your desired number of clusters.

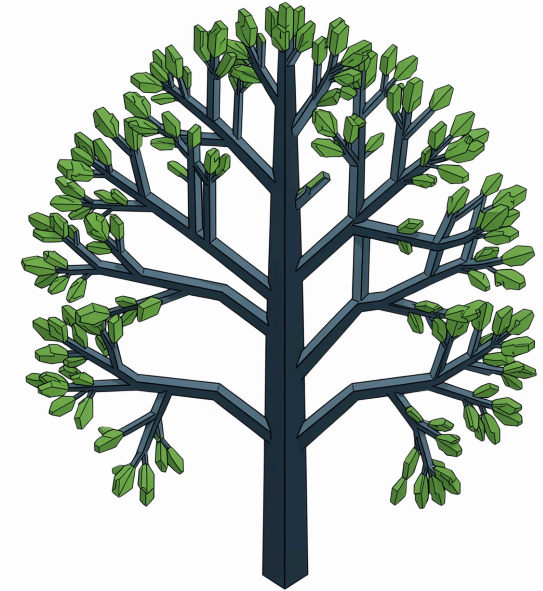
Worked Example — Agglomerative Clustering

Consider 5 students: **A, B, C, D, E** grouped by exam scores. Distance = difference in marks.



📄  **Cut at Step 4** $\rightarrow$ 2 clusters: $\{A,B,C\}$ and $\{D,E\}$ — students with similar performance grouped together!

Divisive Clustering — Step by Step (Top-Down)



1 Initialize

Place **ALL** N data points into a single cluster.

2 Select

Pick the cluster with the highest internal dissimilarity.

3 Split

Divide it into two sub-clusters that maximize the distance between them.

4 Repeat

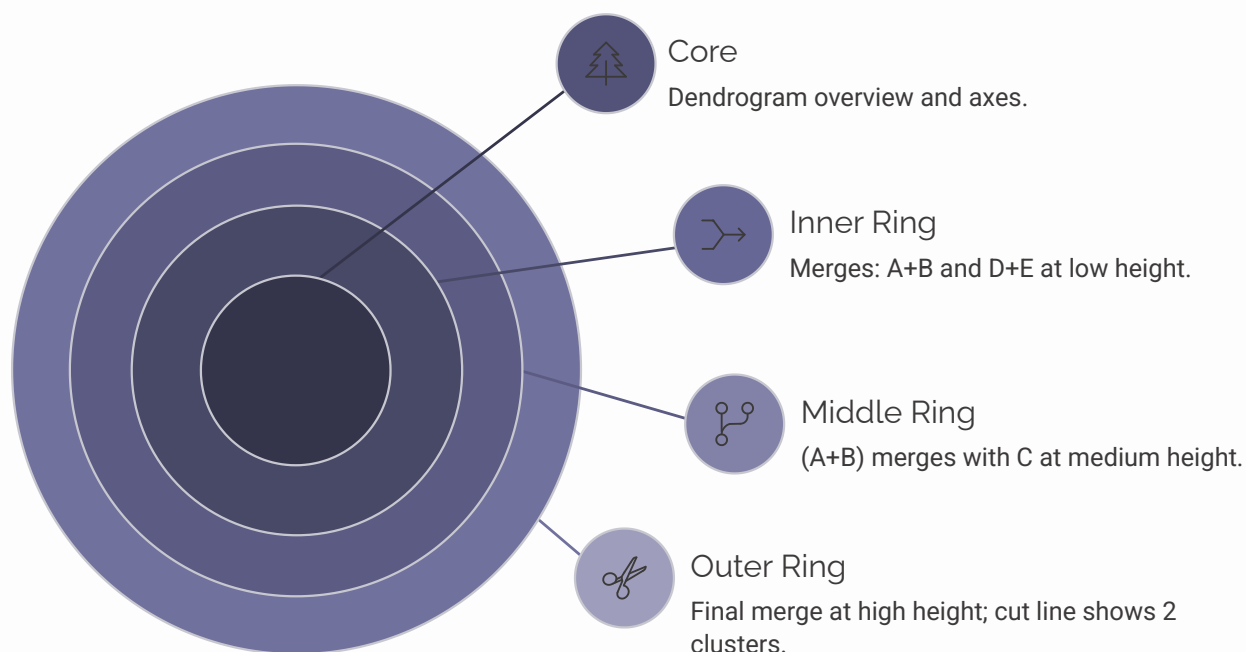
Keep splitting until every point is its own cluster or a stopping criterion is met.



 **Analogy:** Like splitting a country → states → districts → cities — always dividing from the top down.

Divisive methods see the "big picture" first; Agglomerative builds from individual details. Divisive is less common but useful when the goal is to identify large, distinct clusters first.

The Dendrogram — Reading the Tree



What it is

A tree diagram where **leaves** = **individual data points** and **branches** = **merges or splits**. Builds nested cluster relationships without requiring you to specify the number of clusters in advance.

Y-axis (Height)

Represents the **distance/dissimilarity** at which two clusters merged – a higher merge means the groups are more different from each other.

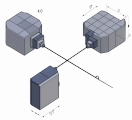
How to Cut

Draw a horizontal line across the dendrogram → the number of vertical lines it crosses = number of clusters you get.

The Power

One dendrogram → infinite clustering options. Cut high for few clusters, cut low for many – different heights reveal different granularities of the same structure.

Linkage Criteria — How to Measure Distance Between Clusters



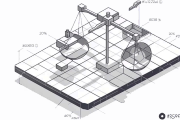
Single Linkage (MIN)

Distance = **closest pair** of points between two clusters.
Tends to create long, chain-like clusters. Sensitive to outliers.



Complete Linkage (MAX)

Distance = **farthest pair** of points between clusters.
Creates compact, tight, roughly equal-sized clusters.



Average Linkage

Distance = **average of all pairwise distances** between two clusters. A balanced approach — less sensitive to outliers.



Ward's Method

Minimizes **total within-cluster variance** after merging. The most popular choice for real-world applications — produces well-separated clusters.

Real-World Applications



Biology & Bioinformatics

Grouping genes with similar expression patterns; building evolutionary trees (phylogenetics) showing species relationships.



Document Clustering

Grouping news articles by topic. Same dendrogram – cut at different levels for broad or specific topics. One run, many insights.



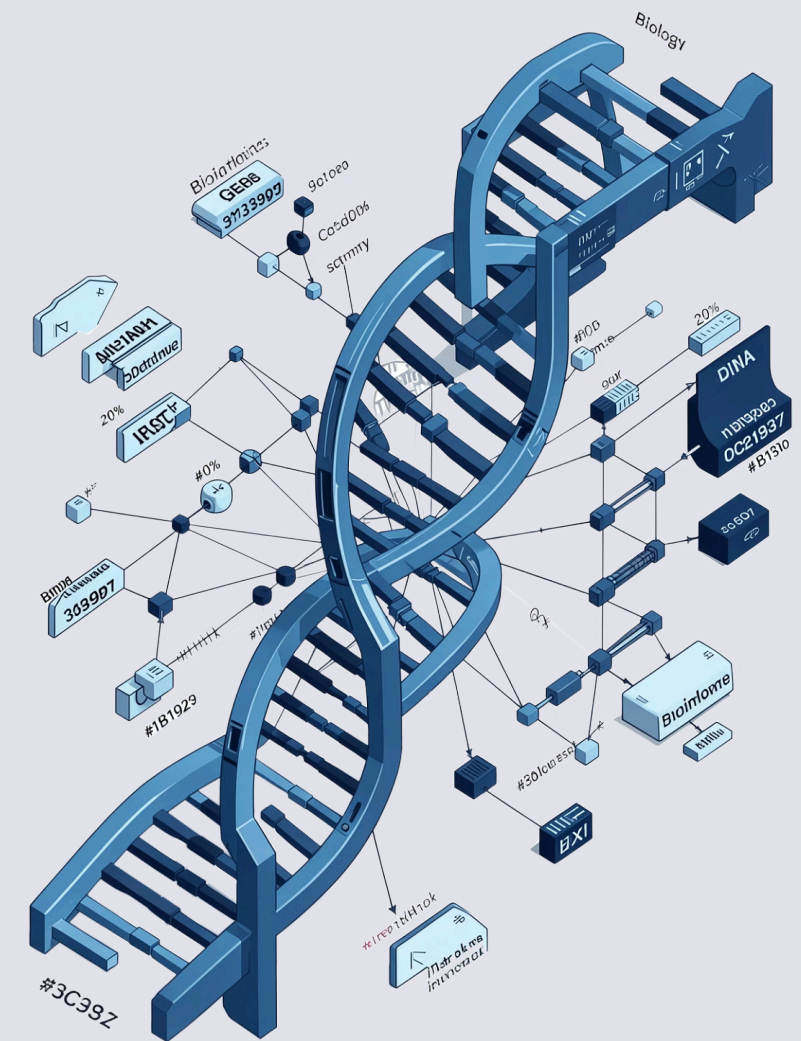
Customer Segmentation

Group customers by purchase behavior. A marketing analyst can cut the dendrogram at 5 clusters for broad strategy or 20 for precision campaigns.



Medical Diagnosis

Cluster patients by symptoms to identify disease subtypes and personalize treatment plans at scale.



Key Takeaways & Quick Revision

→ Hierarchical Clustering

Builds a tree of clusters — **no need to pre-specify K**. One algorithm run gives you every possible clustering at every granularity.

→ Agglomerative (Bottom-Up)

Merge → Merge → Merge until 1 cluster. **Most commonly used** in practice. Complexity: $O(n^2)$.

→ Divisive (Top-Down)

Split → Split → Split until N clusters. **Sees big picture first**. More computationally expensive: $O(2^n)$.

→ Dendrogram

The visual output — cut it at any height to get your desired number of clusters. Y-axis = distance between merged groups.

→ Linkage Methods

Single, Complete, Average, **Ward's** — controls how inter-cluster distance is measured. Ward's is most popular for real-world use.



Remember:

"Agglomerative = ants building a colony from scratch.

Divisive = breaking a pizza into slices!"

References

Hierarchical Clustering: Complete Guide with Dendrograms, Linkage Criteria & Implementation – Michael Brenndoerfer

Hierarchical Clustering in Python – Complete Guide | W3docs

Hierarchical Clustering – Wikipedia

Hierarchical Clustering Explained: Building and Reading Dendrograms in Python – Techietory

Thank You

This concludes the session on **Hierarchical Clustering**. We covered the core concepts, two algorithmic approaches, a hands-on worked example, dendrogram interpretation, linkage criteria, and real-world applications.

Questions?

Discuss with your classmates and try implementing agglomerative clustering in Python using `scipy.cluster.hierarchy`.

Next Steps

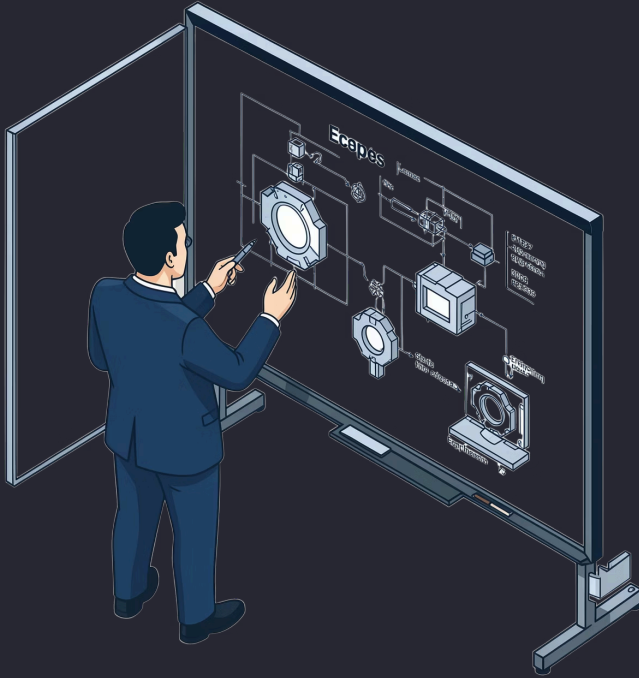
Explore K-Means, DBSCAN, and compare their trade-offs against Hierarchical Clustering in your lab assignments.

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About the Instructor



Dr. Anilkumar V. Brahmane

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Dr. Brahmane specializes in **Machine Learning, Data Mining, and Artificial Intelligence** for undergraduate and postgraduate engineering students.

- This presentation is designed for B.Tech students learning clustering algorithms as part of their Machine Learning or Data Mining curriculum.

Department

Computer Engineering

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Focus Area

ML, Data Mining & AI Education